SYSTEM OVERVIEW / SIGNAL FLOW

USB-C PD and Pi host → hardware interlock → transformer drive → HV multiplier → sensing

01_power_mcu

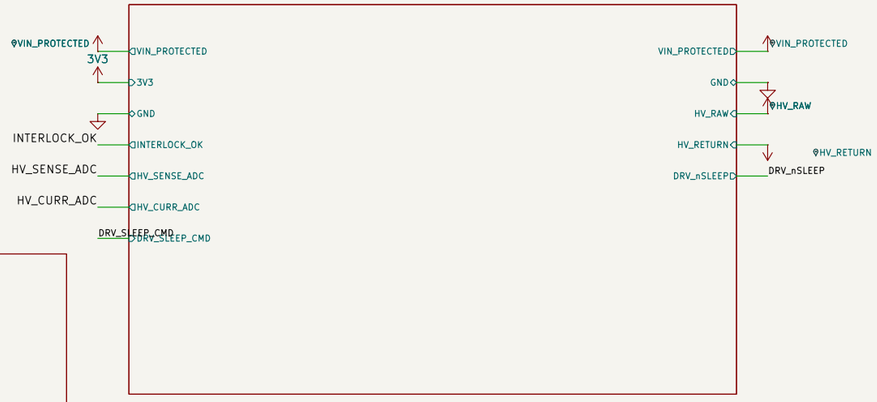


Pi interface and programmable input

05_illumination

J50 <-> J51 / 4-wire cable

04_sense_safety



Voltage/current sensing and hardware interlock

02_drive_transformer



PWM bridge and resonant transformer

03_multiplier



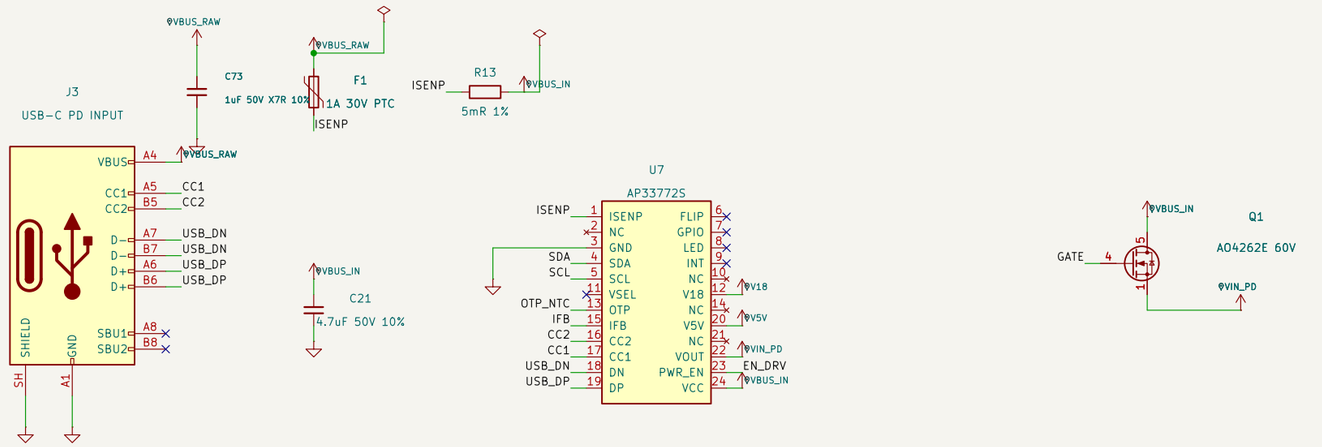
High-voltage ladder and passive discharge

Antigravity Engineering	
Sheet: /	
File: pi5_corona_hv.kicad_sch	
Title: Raspberry Pi 5 HAT Corona HV Supply – System Overview	
Size: A3	Date: 2026-10-03
KiCad E.D.A. 10.0.6	Id: 1/6

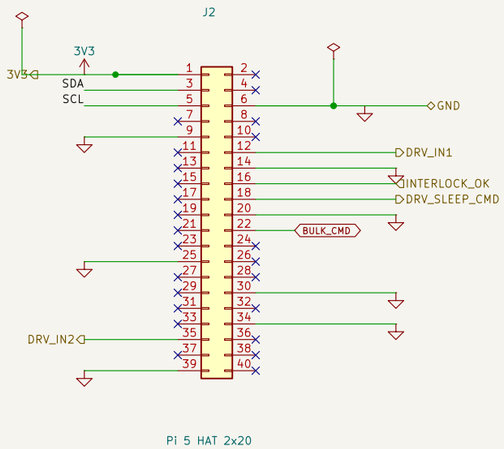
Rev: K

PI 5 HAT / USB-C PD / ENVIRONMENTAL AND OPTICAL SENSING

USB-C INPUT AND PROGRAMMABLE PPS SINK

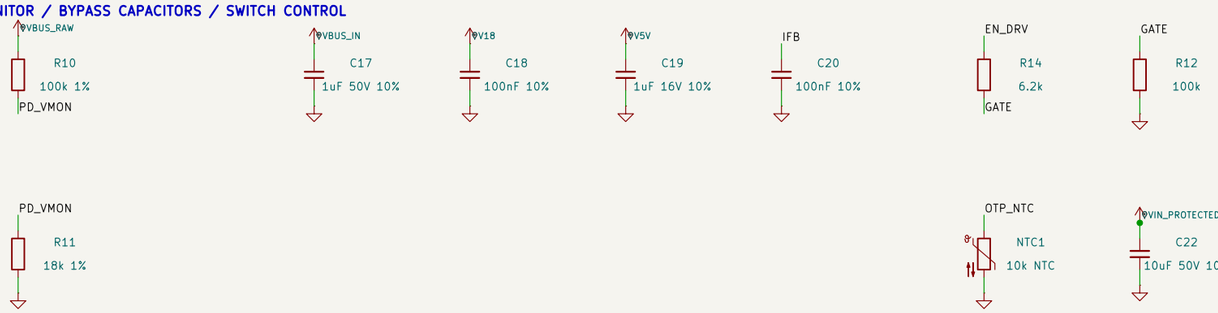


RASPBERRY PI 5 / 40-PIN HOST INTERFACE

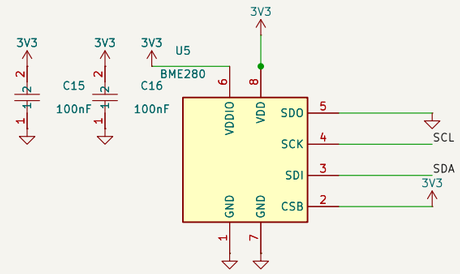


Pi 5 HAT 2x20

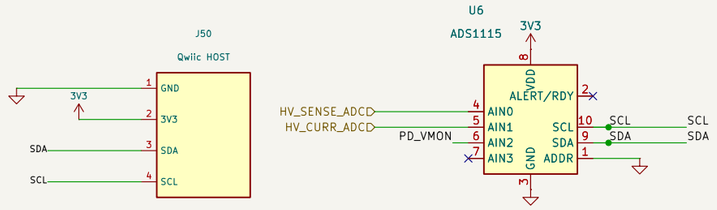
INPUT MONITOR / BYPASS CAPACITORS / SWITCH CONTROL



BME280 0x76 / ADS1115 0x48 / J50 REMOTE LIGHT SENSOR

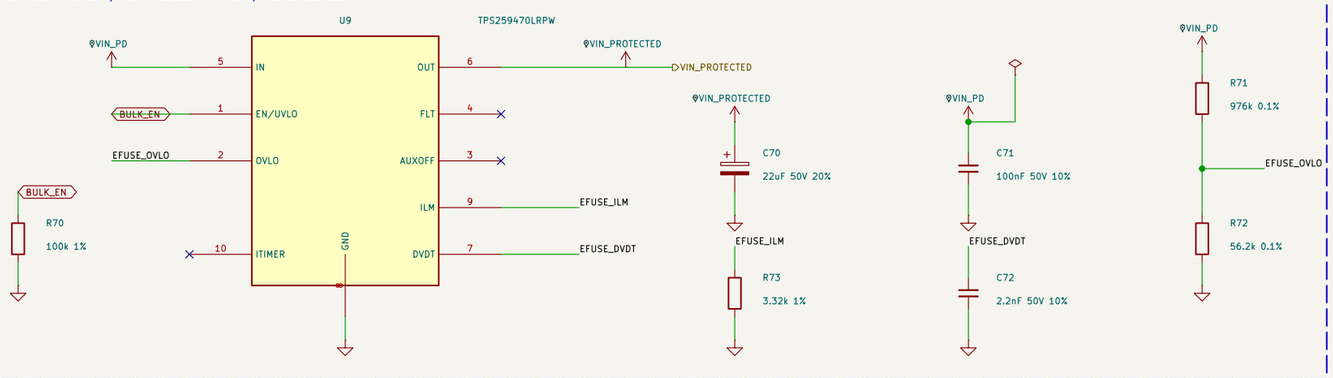


ADC: AIN0 HV voltage / AIN1 HV current / AIN2 VBUS / AIN3 spare.



J50 -> J51: Qwiic / STEMMA QT 3.3 V cable to break-off illumination board.

POST-CONTRACT BULK POWER / CONTROLLED INRUSH / REVERSE BLOCKING



5-20 V PPS requests require a compatible USB-C PD source. The sink starts at 5 V.

DRV8876 bridge: require >= 6 V PD request before HV enable; verify local VM >= 4.5 V under load.

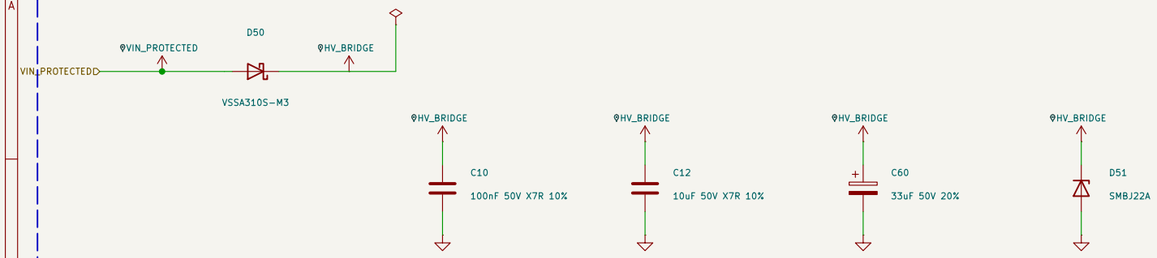
GPIO25: BULK_EN only after explicit PD contract; disable on detach/reset/fault. Request >= 6 V for HV.

GPIO18/19: PWM0/1. GPIO24: sleep. GPIO23: interlock. U9 limits inrush; capacitor budget includes all rails.

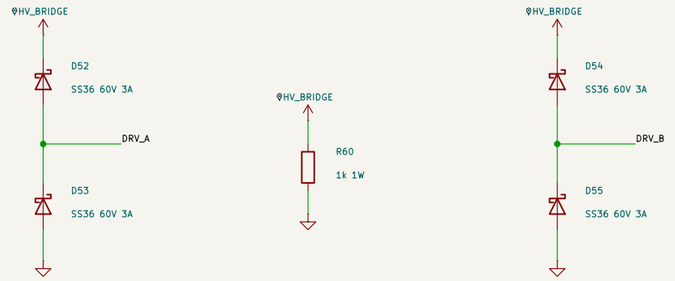
Antigravity Engineering	
Sheet: /01_power_mcu/	
File: 01_power_mcu.kicad_sch	
Title: 01 Power Input, Raspberry Pi 5 HAT & Sensors	
Size: A3 | Date: 2026-10-03	**Rev: K**
KiCad E.D.A. 10.0.6	Id: 2/6

FULL BRIDGE WITH INDUCTIVE TRANSIENT PROTECTION

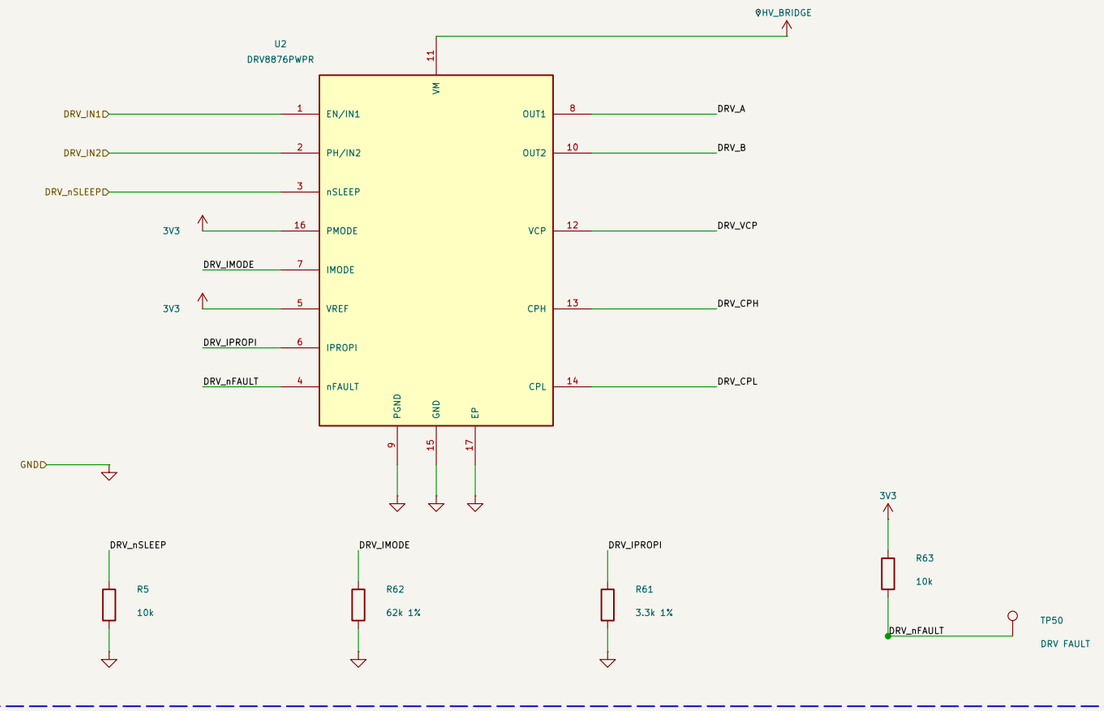
INTERLOCKED INPUT / REGENERATION ENERGY CLAMP



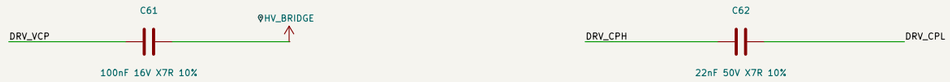
OUTPUT RECIRCULATION CLAMPS / RESERVOIR BLEEDER



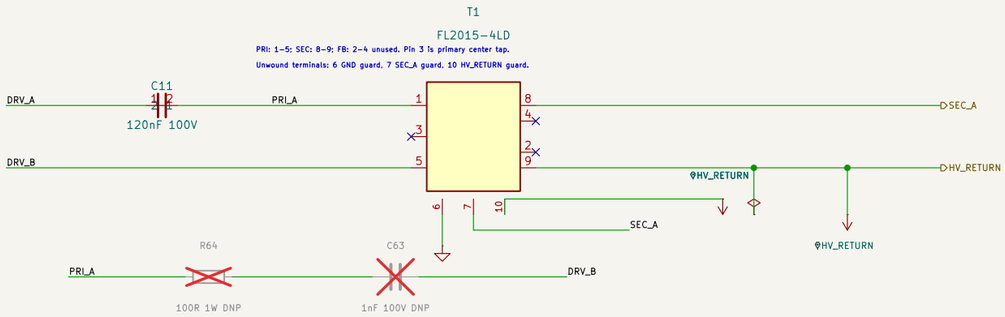
PWM BRIDGE / CURRENT LIMIT / FAULT TEST POINT



CHARGE PUMP DECOUPLING



RESONANT TRANSFORMER / OPTIONAL PRIMARY SNUBBER



PMODE = 3V3: PWM mode. R61 sets -1.0 A trip; R62 selects cycle-by-cycle / latched OCP.

D50 isolates returned energy; D51 clamps VM. C60 = 33uF to keep total PD sink capacitance below 100uF.

Require >= 6 V requested input before HV enable; verify local VM >= 4.5 V under load. PWM <= 100 kHz.

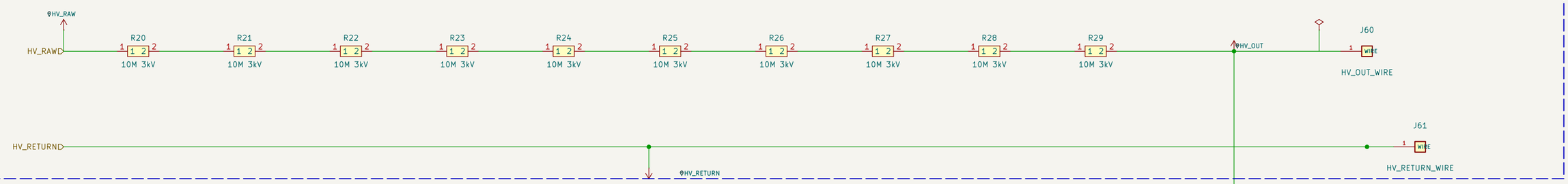
R64/C63 DNP: tune from measured primary ringing. Verify VM/OUT spikes, TVS energy and temperature on PCB.

K1: regulated 3V coil; independent relay contacts gate BULK_CMD and DRV_SLEEP_CMD. GPIO23 senses coil feed through R92.

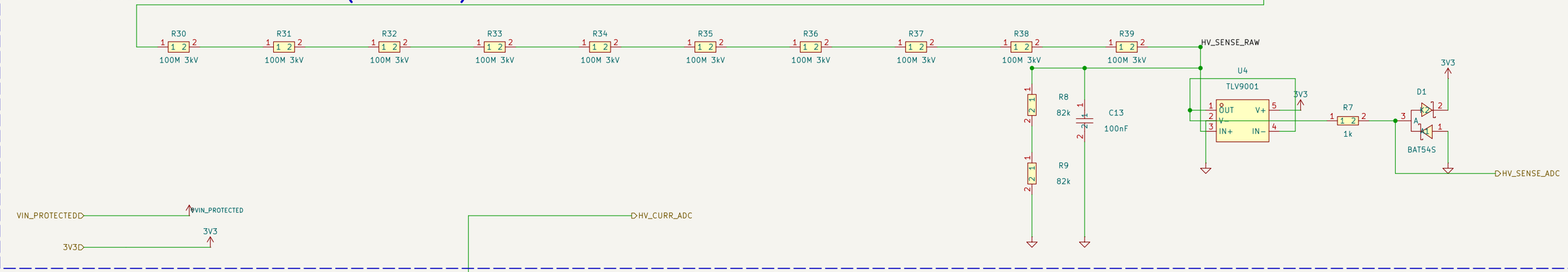
Antigravity Engineering	
Sheet: /02_drive_transformer/	
File: 02_drive_transformer.kicad_sch	
Title: 02 Isolated AC Generation & Resonance Tank	
Size: A3	Date: 2026-10-03
KiCad E.D.A. 10.0.6	**Rev: K**
	Id: 3/6

VOLTAGE / CURRENT SENSING / HARDWARE INTERLOCK

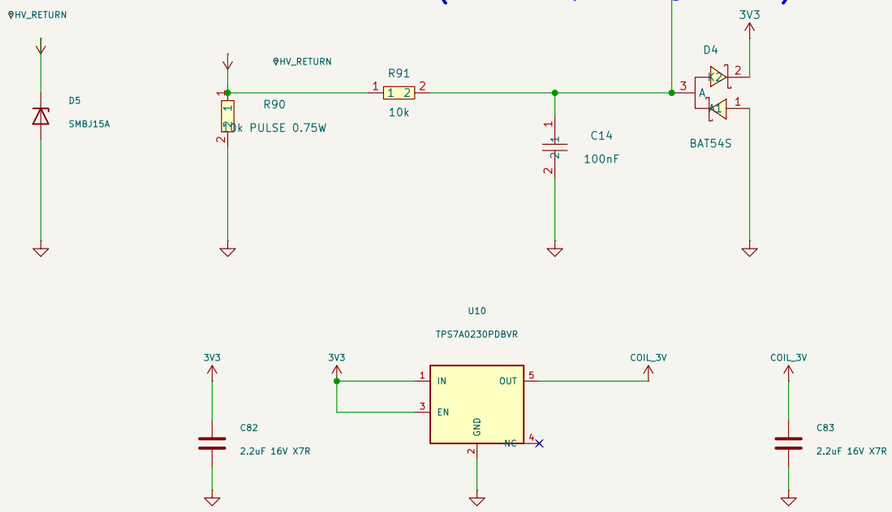
OUTPUT BALLAST STRING (100 M-OHM TOTAL, 150 uA SHORT-CIRCUIT LIMIT)



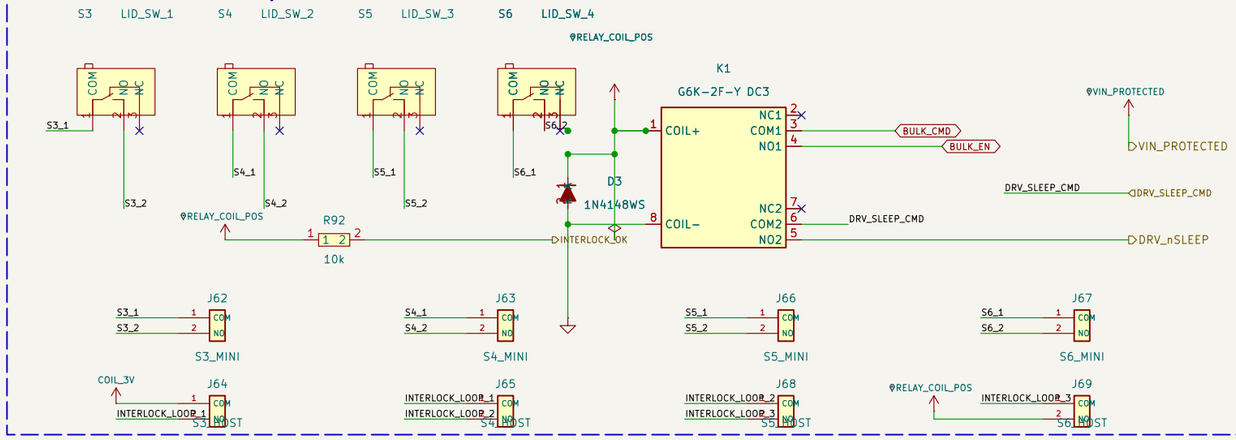
1 G-OHM HIGH VOLTAGE DIVIDER (RATIO 6098:1) & BUFFER



HV RETURN CURRENT SHUNT (10 k-OHM, 1.5 V @ 150 uA)



SAFETY INTERLOCK / DETACHABLE SWITCH BOARDS



Detachable switch boards (S3-S6) wire to host headers (J64, J65, J68, J69). Series interlock loop on host. Pin 1 COM, pin 2 NO.

D5: 15V standoff / 24.4V clamp; 1uA max leakage at 25C. R90 pulse-rated; R91 limits ADC-clamp current.

Antigravity Engineering

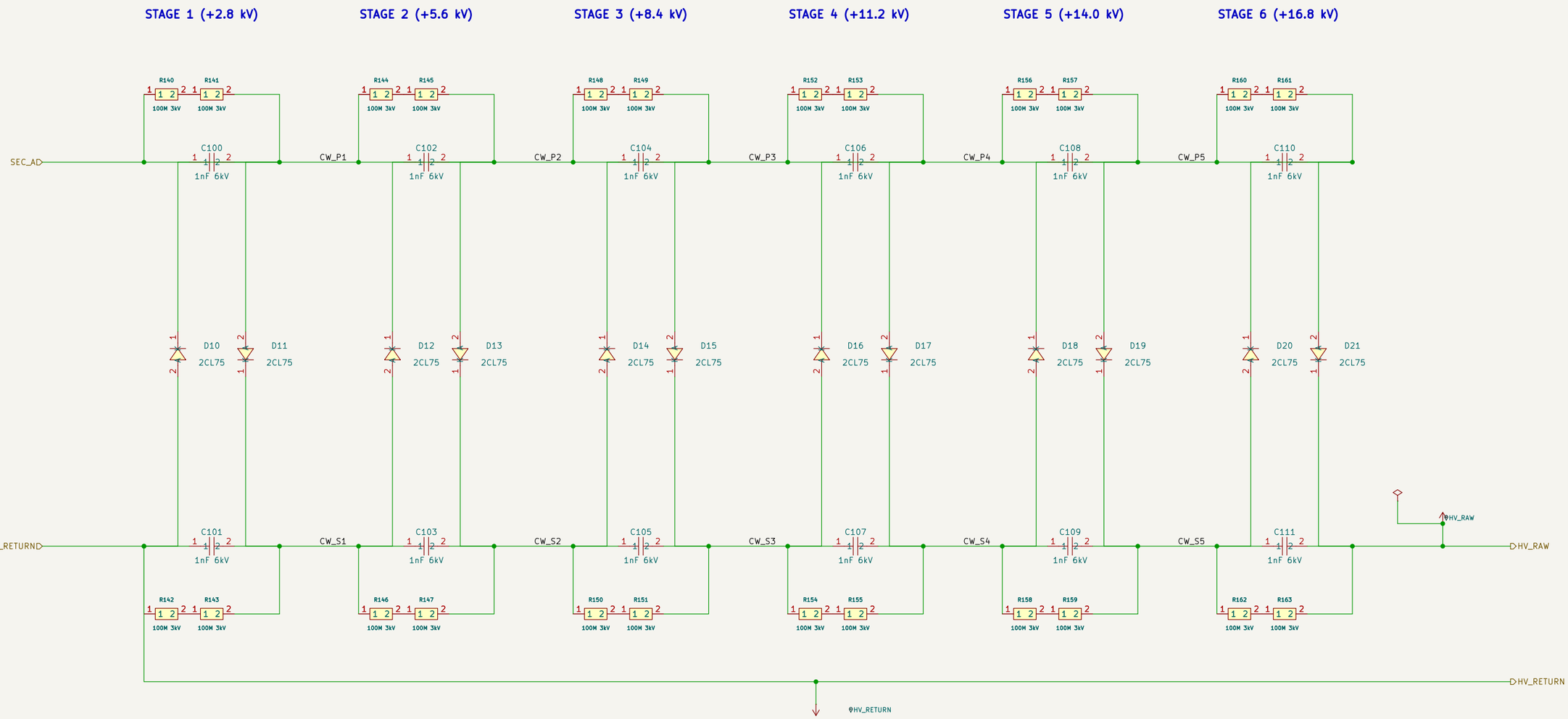
Sheet: /04_sense_safety/

File: 04_sense_safety.kicad_sch

Title: 04 Ballast, Divider, Sensing & Hardware Interlock

Size: A3	Date: 2026-10-03	Rev: K
KiCad E.D.A. 10.0.6		Id: 4/6

6-STAGE POSITIVE COCKCROFT-WALTON VOLTAGE MULTIPLIER & PASSIVE BLEEDERS

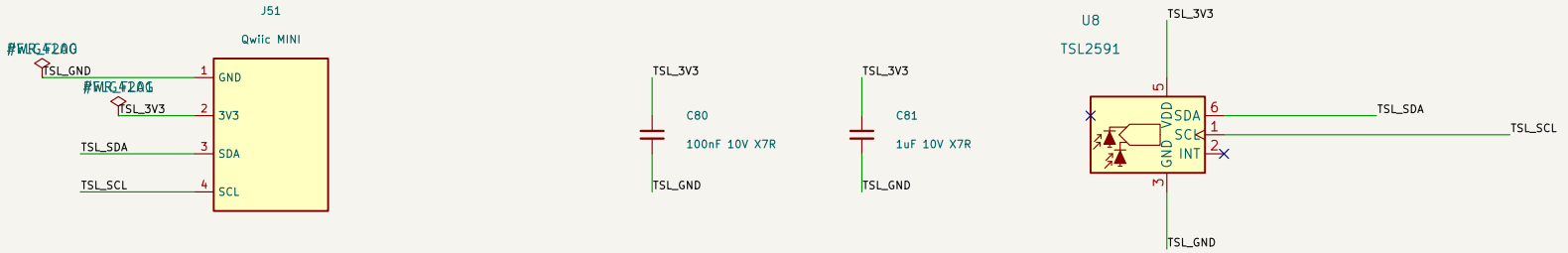


SAFETY & DISCHARGE SPECIFICATIONS:

1. Nominal open-circuit output voltage: ~17 kV DC (regulated by MCU PWM duty cycle to +10 kV .. +15 kV DC).
2. BLEEDERS: 12 banks x 2 series 100M / 3kV 2512; each bank is 200M. Verify <50V discharge on assembled hardware.
3. CLEARANCE: Minimum 4.0 mm clearance maintained between adjacent stages; all inner copper planes removed beneath ladder.

Antigravity Engineering		
Sheet: /03_multiplier/		
File: 03_multiplier.kicad_sch		
Title: 03 6-Stage Positive Cockcroft-Walton Multiplier & Bleeders		
Size: A3	Date: 2026-10-03	**Rev: K**
KiCad E.D.A. 10.0.6		Id: 5/6

BREAK-OFF OPTICAL MINI BOARD / TSL2591 0x29



J50 on HAT; J51 on mini board. Cable pin 1 GND, 2 3V3, 3 SDA, 4 SCL.

Route only these four nets across the scored break-off neck; cutting the neck severs all four.

After detaching, a 1:1 Qwiic / STEMMA QT cable restores all four connections. 3.3 V only.

Keep C80/C81 beside U8. No local pull-ups or LEDs; use the Pi bus pull-ups.

Keep the optical aperture and copper neck away from high-voltage circuitry; start cable operation at 100 kHz.

Antigravity Engineering		
Sheet: /05_illumination/		
File: 05_illumination.kicad_sch		
Title: 05 Illumination Break-off Mini Board		
Size: A4	Date: 2026-10-03	**Rev: K**
KiCad E.D.A. 10.0.6		Id: 6/6